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Visual Perception Challenges in Adverse Weather for Autonomous Vehicles: A Review of Rain and Fog Impacts

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***Abstract*—Autonomous driving systems (ADS) offer efficient solutions for transportation, but their visual perception capabilities are significantly affected under adverse weather conditions such as rain and fog, posing a threat to vehicle safety. Rain droplets and fog can interfere with sensors, leading to inaccurate information acquisition and increased driving risks. Therefore, enhancing the environmental perception of ADS in adverse weather conditions is a critical research focus. This paper systematically evaluates the impacts and challenges of adverse weather conditions, specifically rain and fog, on ADS environmental perception sensors. It explores the imaging characteristics of rain and fog and their effects on visual perception, summarizing dehazing and deraining algorithms. Additionally, this paper reviews the development status of advanced visual tasks such as object detection in autonomous driving under adverse weather conditions. The aim of this paper is to provide a comprehensive understanding of the reliability of ADS in adverse weather conditions, laying a foundation for future developments and practical applications in autonomous driving technology.**

***Index Terms*—Adverse Weather, Environmental Perception, Autonomous Vehicles, Advanced Computer Vision Tasks.**

I. Introduction

Autonomous driving relies heavily on robust perception capabilities, requiring accurate understanding and interpretation of the vehicle's surrounding environment. Significant advancements have been made in the field of environmental perception, enabling autonomous vehicles to operate smoothly under favorable weather conditions. However, challenges persist under adverse weather conditions. If the perception system degrades due to inclement weather, the decision-making and control abilities of autonomous vehicles may be compromised. Common weather phenomena such as rain, snow, or fog can affect sensor data, reducing the signal-to-noise ratio and significantly increasing environmental noise [1].

In adverse weather conditions such as rain and fog, improving visual and environmental perception capabilities plays a crucial role in the field of autonomous driving. However, weather conditions directly affect the state of the environment, impairing the perception abilities of autonomous driving system sensors and further complicating perception tasks such as target detection. Changes in environmental conditions can also cause discrepancies between remote sensing results and map information, affecting localization accuracy. Therefore, it is essential to deeply investigate the impact of adverse weather conditions on different sensors and clearly understand how current adverse weather models interfere with environmental perception. This knowledge will help the perception modules of autonomous driving systems to accurately adapt to changing traffic environments, thereby ensuring the safety of autonomous vehicles [2]. However, there

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has not yet been a comprehensive review specifically focused on visual perception and advanced visual tasks for autonomous driving in rainy and foggy weather. In addition to filling the literature gap, the main contributions include:

1.This paper comprehensively summarizes the impact of rain and fog on sensors in autonomous vehicles and provides a detailed analysis of the challenges faced by visual perception in these conditions. By examining the optical imaging model of rain and fog, it also introduces the current state and issues of visual image enhancement algorithms under adverse weather interference.

2.Based on the characteristics of optical image formation in rainy and foggy environments, a detailed comparative analysis of the current shortcomings and solutions in object detection, semantic segmentation, and depth estimation tasks within autonomous driving visual perception algorithms is presented.

3.Prospects for the Development and Future Research Directions of Autonomous Driving Environmental Perception in Adverse Weather Conditions.

II. The Impact of Rain and Fog on Autonomous Vehicle Sensors

Currently, sensors equipped in autonomous vehicles fall into three main categories: vehicle self-perception sensors, vehicle localization sensors, and environmental perception sensors.Environmental perception sensors utilize external sensors such as optical cameras, infrared cameras, LiDAR, radar, and ultrasonic sensors to accurately perceive the vehicle's external environment. Therefore, ensuring high reliability and accuracy of sensors is paramount. However, weather conditions directly impact environmental status, disrupting the perception capabilities of autonomous driving system sensors and further complicating perception tasks such as target detection and object recognition. Changes in environmental conditions also lead to discrepancies between remote sensing results and map information, affecting localization accuracy.

A. Impact of Rain and Fog on LiDAR

LiDAR is an active sensor capable of creating an accurate 3D traffic scene, assisting autonomous vehicles in efficiently completing 3D environmental perception tasks. However, due to light scattering and occlusion, the performance of LiDAR changes under adverse weather conditions such as fog, rain, or snow. According to the study by Fersch et al. [3] on small-aperture LiDAR sensors, normal rainfall has little impact on the functionality of LiDAR. The impact of rainfall on LiDAR may not only involve a reduction in signal power but also affect the accuracy and completeness of the point cloud, which is difficult to assess through mathematical models or simulations. While light drizzle has minimal effect on LiDAR, increased rainfall rates may compromise LiDAR accuracy. Uneven precipitation in the atmosphere is likely to form mist clusters, causing erroneous detections by LiDAR. Therefore, in heavy rain, it can be treated as dense fog or thick smoke. Hasirlioglu et al. [4] demonstrated through simulations using a layered approach to signal transmission paths that signal reflection intensity significantly decreases at precipitation rates of 40 mm/h and 95 mm/h. This model was validated using a laser rangefinder in a handmade rain simulator. The LIBRE dataset collected by Carballo and Lambert et al. [5] shows significant loss and noise interference in the LiDAR point cloud signals in Fig 1 due to the influence of rain and fog weather conditions. LiDAR signals emitted at 905nm and 1550nm wavelengths are subject to scattering effects from raindrops.

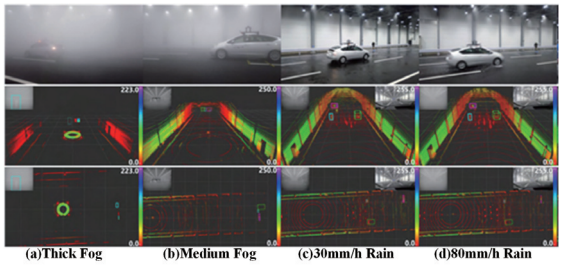

Fig. 1. Point Cloud Visualization Comparison of LiDAR under Different Levels of Rain and Fog.

B. Impact of Rain and Fog on Vision Sensors

In rainy and foggy environments, the scattering effect of raindrops on light causes deflection and

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diffusion of light rays, reducing the energy density of reflected light. This leads to image blurring and decreased contrast. Water droplets and suspended particles in fog can absorb part of the light, especially within the visible spectrum, further weakening light intensity. The uneven propagation of light in rain and fog can cause uneven illumination in images, increasing image noise. Additionally, raindrops and fog can cause refraction and perspective distortion of light rays, leading to distortion of the shape and position of objects in images.

Furthermore, rain and fog significantly reduce visibility, making objects in visual images appear blurry and severely affecting the representation of color and feature information of detected objects. This greatly distorts image features and reduces the performance of deep learning-based object detection and recognition. Decreased visibility also impacts the camera's perception of scene depth, making it difficult to identify distant objects. In rainy and foggy driving environments, due to the presence of raindrops and fog, vision sensors may require longer exposure times to capture sufficient light, leading to a decline in real-time performance. Consequently, targets in image sequences may exhibit unrealistic motion estimation, affecting the analysis of dynamic scenes . Fig 2 shows a comparison of the impact of real traffic environments in rain and fog on the environmental perception of vision sensors.

III. The Impact of Rain and Fog on Visual Perception

A. Importance of Visual Environment Perception

AI-based visible light camera vision perception methods have become the dominant solution in the field of autonomous driving environment perception due to their affordability and intuitiveness. However, adverse traffic conditions such as rain, fog, and low light can cause visual perception systems to malfunction, reducing the perception capabilities of autonomous vehicles and potentially leading to traffic safety incidents. As shown in Fig 2, visual perception algorithms exhibit lower detection performance in challenging outdoor environments such as fog and rain.

(a)Clear weather conditions (b)Heavy fog traffic conditions (c)Rainy traffic conditions

Fig. 2. Comparison of the Impact of Rain and Fog on Visual Sensor Perception (Object Detection) in Real Traffic Environments.

Visual perception systems in rainy and foggy environments remain one of the challenging tasks in the field of autonomous driving. The main issues include the following aspects: (1) How to enhance the robustness of perception algorithms under adverse weather conditions like rain and fog; (2) How to obtain the datasets required for testing models in rain and fog traffic scenarios; (3) How to quantify the performance of deep learning-based vision systems in adverse scenarios such as rain and fog.

B. Optical Image Formation Model in Rain and Fog

To accurately describe the formation process of foggy images, McCartney first proposed the atmospheric scattering model [6]. This model explains that fog is produced when a large amount of water vapor and suspended tiny particles in the atmosphere absorb and scatter natural light. The mathematical model for imaging on a foggy day is:

$$I(x) = D + A = J(x)\,t(x) + A\,(1 - t(x)), \quad (1)$$

where I represents the foggy image obtained by the detection system,J denotes the clear image that needs to be restored, x indicates the position of the pixel in the image, A denotes the atmospheric light value from an infinitely distant source, and $t(x)$ represents the transmission function, which physically signifies the proportion of light that can reach the detection system after being attenuated by particles.

The simplest and most commonly used composite model is:$\mathbf{O} = \mathbf{B} + \mathbf{S}$,where $\mathbf{B}$ is the background layer, $\mathbf{S}$ is the rain streak layer, and $\mathbf{O}$ is the image degraded by the rain streaks. However, due to the

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uncertainty of rain streaks in real environments, which can create visual effects of rain accumulation and haze, Li et al. [7] proposed a more realistic rainy-day imaging model. The mathematical model can be expressed as:

$$\mathbf{O} = \mathbf{T} \odot \left(\mathbf{B} + \sum_{i}^{n} \widetilde{\mathbf{S}}_i \right) + (\mathbf{1} - \mathbf{T}) \odot \mathbf{A}, \quad (2)$$

where $\widetilde{\mathbf{S}}$ represents the rain streak layer with raindrops falling in the same direction, $n$is the maximum number of rain streak layers, and $\mathbf{T}$ is the transmission map of scattering particles.

C. Image Dehazing and Deraining

To reduce the impact of haze on image data and improve the accuracy of target detection in intelligent driving vehicles, image dehazing is a crucial step in the post-processing phase of target detection. Currently, image dehazing algorithms can be categorized into four types: Image Enhancement-Based Methods: These methods typically use traditional image processing techniques to enhance the contrast, brightness, and gradient features of the image. However, since they do not address the dehazing based on the mechanism of fog formation, the resulting images often suffer from color distortion and loss of details. Physical Model-Based Methods: These methods establish a degradation physical model of fog based on the scattering mechanism of atmospheric light and use prior knowledge of the image, such as the saturation prior, sparsity prior, and regularized probability prior, to achieve dehazing. For example, He et al. [8] proposed the dark channel prior, which assumes that in the clear image's object background regions, there exists a color channel with very low values. Using this prior knowledge, dehazing is performed based on the atmospheric scattering model. Neural Network and Deep Learning-Based Methods: These methods can be divided into two types: Model-Based Network Architectures: For instance, Li et al. [9] proposed the AOD-Net and Cai et al. [10] proposed the DehazeNet based on CNN, which heavily rely on the atmospheric scattering model. End-to-End Direct Dehazing Methods: For example, Liu et al. [11] proposed a grid-shaped dehazing network that effectively mitigates the bottleneck problem in traditional multi-scale fusion dehazing methods through an attention mechanism. However, these end-to-end methods often fail to fully extract and utilize multi-scale feature information at different resolution levels and to effectively extract and fuse features from different spatial contextual levels during the feature transformation stage. This limitation in feature receptive field size results in incomplete dehazing, leaving residual fog in the image. Fig 3 shows the visual effects of several classic image dehazing algorithms in real traffic environments.

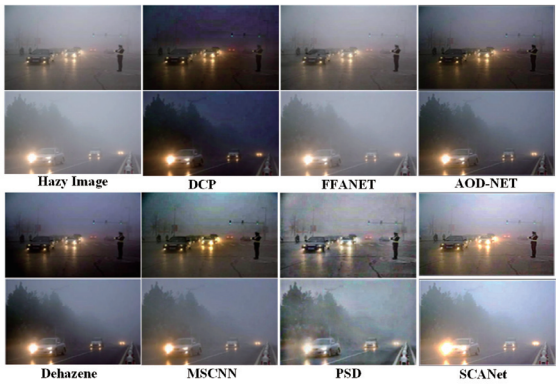

Fig. 3. Visualization of Image Defogging in Real Traffic Environments.

Current image deraining methods are mainly divided into two categories: model-based deraining methods and data-driven deraining methods. As early as 2004, studied image deraining algorithms and proposed a video deraining method based on the motion characteristics of rain streaks. Chen et al. [12] used signal separation techniques to separate rain streaks from the background layer for image deraining. Li et al. [13] estimated multi-scale rain streak information and used a Gaussian mixture model for image deraining. With the rapid development of convolutional neural networks in the field of image processing, since 2017, image deraining has shifted from model-driven approaches to the data-driven era. Deep convolutional neural networks automatically extract multi-level image features and learn more accurate deraining mapping functions for precise image deraining. Although

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deep learning-based methods significantly outperform model-based methods in deraining effectiveness, convolutional kernels only focus on limited local features at the pixel level, failing to effectively unify global features and contextual information of image data. With the rapid rise of Transformers in the field of image processing, self-attention (SA) modules are used as the backbone network to model non-local information for the deraining task. For example, in the image restoration field, models such as IDT [14], Restormer [15], Uformer [16], and DRSformer [17] not only eliminate other noise in images but also remove rain and fog interference to some extent. Fig4 shows the visual effects of classic image deraining algorithms in real traffic environments.

Fig. 4. Visualization of Image Deraining in Real Traffic Environments.

IV. Visual Perception Algorithms for Autonomous Vehicle in Rainy and Foggy

A. Object Detection in Rainy and Foggy

To effectively address the decline in object detection performance due to weather conditions in autonomous driving scenarios, current approaches often employ data-driven methods to mitigate domain shift issues in real-world scenarios. However, due to the inherent variability of adverse weather environments and the lack of training images from real-world scenarios, although synthetic datasets have shown some effectiveness in specific scenes, detection efficiency may drastically decrease with slight variations in external environmental conditions. Chen et al. [18] addressed domain adaptation issues through domain adaptive methods, using internal image features to reduce domain shift and enhance object detection accuracy. Nevertheless, this method requires computation of extensive deep-level image features, resulting in slower detection speeds.

Some prior-based methods combine image enhancement and object detection to mitigate interference from adverse weather-specific information. Singadi et al. [19] utilized prior knowledge of adverse weather imaging, employing adversarial pre-loss with feature recovery to mitigate adverse weather effects on object detection. However, this method is effective only for specific weather conditions.To minimize domain shift under different weather conditions, Li et al. [20] proposed a novel domain adaptation framework to achieve robust detection performance of autonomous driving targets under heavy fog conditions. Liu et al. [21] introduced IA-Yolo, utilizing a small CNN network to predict differentiable image processing IP parameters and an image dehazing module to enhance foggy images, integrated with Yolov3 for end-to-end object detection. Fig2 illustrates the visual effects of these classic object detection methods under adverse conditions such as rain and fog.

B. Semantic Segmentation in Rainy and Foggy

Semantic segmentation is a critical module in autonomous driving perception and recognition, with current research primarily focusing on improving accuracy in good weather or low-light conditions, rather than robustness, scalability, and computational efficiency in adverse rain and foggy environments. Deep learning techniques automatically learn high-level feature representations of images, enabling more accurate identification and segmentation of different semantic regions in images. Most deep learning-based semantic segmentation methods are supervised learning models. Fully supervised image semantic segmentation methods use manually annotated pixels as training samples, facilitating efficient and accurate network training.

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Semantic segmentation in rainy and foggy environments has long been a research challenge in autonomous driving visual perception. Currently, the field faces numerous issues and challenges. Firstly, raindrop occlusion and scattering effects, along with foggy blurring, degrade image quality, making accurate detection and segmentation of targets difficult. Secondly, complex background interference in rainy and foggy environments further complicates segmentation algorithms. Additionally, the lack of large-scale annotated data limits the training and generalization capabilities of algorithms.

C. *Depth Estimation in Rainy and Foggy*

Depth estimation has always been a core challenge in the field of autonomous driving perception, serving as a fundamental component of visual understanding for tasks such as image interpretation, object recognition, and decision-making in traffic and road scenes. However, in rainy conditions, raindrops severely interfere with monocular depth estimation tasks. The rain streaks cause dense occlusions and are close to the camera, resulting in depth maps in rainy scenes that are not only unclear at the edges but also incorrectly position distant objects as nearby. Currently, deep learning-based depth estimation algorithms perform poorly in rainy and foggy conditions. Research has shown that many deep learning-based depth estimation algorithms, similar to other image tasks, modify the network's receptive field by controlling standard convolution strides or pooling operations. While this approach achieves good results for coarse-grained visual tasks like image segmentation and object recognition, precise pixel-level estimation required by depth estimation algorithms is compromised by operations that increase stride or pool size, leading to significant feature loss. In rainy scenarios where rain streaks of different depths have similar colors, extensive use of max pooling operations in depth estimation networks results in localized loss of depth information, causing blurriness at the edges of various objects in the scene depth map, as depicted in Fig5. Addressing complex scenarios like rain and fog in depth estimation can be approached by constructing encoder-decoder networks with visual Transformer structures to extract features at different scales and share information across different levels and decoder parts through skip connections, thereby facilitating long-range context information sharing.

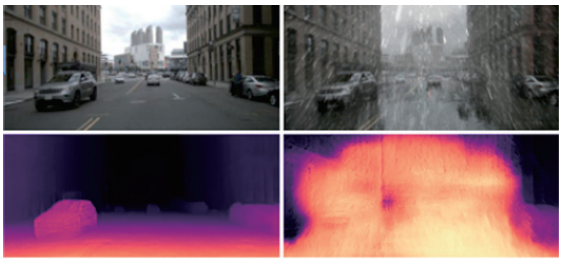

Fig. 5. Depth Estimation Comparison in Rainy and Foggy Conditions (left: clear weather, right: rainy and foggy conditions).

V. Future Research Directions

1.Efficient Image Enhancement Algorithms (Image Deraining and Image Defogging): These are fundamental for improving advanced visual tasks. To enhance the effectiveness of deraining and defogging algorithms, a combined image enhancement network that integrates the strong local feature capabilities of CNN models with the global feature representation abilities of Transformers can be developed. 2.High-robustness visual perception algorithms are crucial for the safe operation of autonomous vehicles. To enhance the environmental adaptability of autonomous vehicles, image enhancement can be combined with visual perception algorithms to improve the domain adaptation capability of tasks such as object detection, semantic segmentation, and depth estimation. This approach enhances the robustness of visual perception in rainy and foggy environments. 3.Efficient Perception Systems: These are essential for the decision-making and control systems of autonomous driving. The overall performance of vehicle perception systems can be improved by optimizing data processing methods, refining multi-sensor fusion strategies, supplementing data signal sources, and enhancing system control capabilities.

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VI. Conclusions

This paper analyzes the impact of adverse weather conditions like rain and fog on common autonomous vehicle sensors, explains the formation mechanisms of rain and fog images, and discusses the issues and solutions of visual image enhancement algorithms. It reviews the current state of object detection, semantic segmentation, and depth estimation algorithms based on visual perception in rainy and foggy environments, and discusses development trends and future research directions for visual perception in autonomous driving.Future research should focus on improving the quality of rain and fog image datasets, enhancing the efficiency of image enhancement algorithms, increasing the robustness of visual perception algorithms, improving the data acquisition capabilities of visual perception sensors, refining the accuracy of vehicle perception systems, and ensuring the driving safety of autonomous vehicles in adverse weather conditions.

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